Reliable Machine Learning with Unreliable Black Boxes	**Spring 2026**

Lecture 11, part 1: Frequency Estimation with Predictions

Instructors: A. Vijayaraghavan and V. Srinivas — Scribe: Andre Shportko

The presentation covered *Improved Frequency Estimation Algorithms with and without Predictions* [Aamand, Chen, Nguyen, Silwal, and Vakilian, 2023] which improves on the learning-augmented frequency estimation of *Learning Based Frequency Estimation Algorithms* [Hsu, Indyk, Katabi, and Vakilian, 2019] and gives tight bounds for CountMin / CountSketch on Zipfian data. The presentation was given by student Dionysis Arvanitakis.

1 Setup

A stream of updates (i, Δ) acts on $f \in \mathbb{R}^n$. WLOG sort so $f_1 \geq \cdots \geq f_n$, and let $N = \sum_i f_i$. The error metric is the *weighted error*

$$\mathrm{WE} = \tfrac{1}{N} \sum_i f_i \cdot |\tilde{f}_i - f_i|,$$

the expected error under a query drawn from f. Throughout we assume Zipfian frequencies $f_i = 1/i$. B is a memory budget.

2 Background

CountMin / CountSketch. CM uses k hash tables of width B/k and reports $\tilde{f}_i = \min_\ell C[\ell, h_\ell(i)]$. CS adds sign hashes s_ℓ and takes the median; lower variance per row, but estimates can be negative.

[**Hsu, Indyk, Katabi, and Vakilian, 2019**] A learned heavy-hitter oracle: predicted heavies are stored exactly in a separate table; everything else goes into a CM/CS sketch.

3 Main Results

	no predictions	with predictions
CountMin	$\Theta(\log n/B)$	$\Theta(\log^2(n/B)/(B \log n))$
CountSketch	$\Theta(1/B)$	$\Theta(\log(n/B)/(B \log n))$
This paper	$O\left(\frac{\log B + \operatorname{poly} \log \log n}{B \log n}\right)$	$O(1/(B \log n))$

4 The Algorithm (No Predictions)

A CountSketch with B columns has additive noise $\|f_B\|_2/\sqrt{B} \approx 1/B$ on Zipfian inputs. So if CS reports an estimate of magnitude $\approx 1/B$ for item i, the signal is below the noise floor: outputting

0 is at least as accurate, and strictly better if $f_i \ll 1/B$.

Construction. Maintain $T = \Theta(\log\log n)$ small CS tables ($S_1, \ldots, S_{T-1}$, each $3 \times B/(6T)$) plus one big one (S_T, $3 \times B/6$). To query i: take the median $\tilde{f}_i$ of the $T-1$ small estimates. If $\tilde{f}_i < O(\log\log n)/B$, return 0; otherwise return the estimate from S_T.

Theorem 1. *For $B \geq \log n$, the expected weighted error on a Zipfian stream is $O\big(\frac{\log B + \text{poly}\log\log n}{B \log n}\big)$.*

5 Learning-Augmented Version

Reserve half the space for exact counts of the predicted top $B/2$ heavy hitters; run the algorithm above on the other half for everything else. Heavy hitters give 0 error, and the tail $i \geq B/2$ is handled exactly as in the case above with B in place of B', yielding $O(1/(B\log n))$.

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Lecture 11, part 2: Discussion: Robustness Auditing

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The second part of class included a discussion about *An Automatic Finite-Sample Robustness Metric* [Broderick, Giordano, and Meager, 2023].

6 Question

Can a study's conclusion be reversed by dropping a tiny fraction of the sample? If so, the result may not generalize.

7 Setup

Data $d_1, \ldots, d_N$, estimator $\hat{\theta}$, scalar quantity of interest $\phi(\theta)$. Reweight data by $\vec{w} \in \{0,1\}^N$; dropping point n means $w_n = 0$. Brute-force enumeration over all subsets is infeasible.

8 AMIP

First-order Taylor expansion in the weights:

$$\phi(\hat{\theta}(\vec{w}), \vec{w}) \approx \hat{\phi} + \sum_n (w_n - 1)\,\psi_n, \qquad \psi_n := \left.\frac{\partial \phi}{\partial w_n}\right|_{\vec{w}=\vec{1}}.$$

The ψ_n are influence scores. To approximate the worst-case drop of $\lfloor \alpha N \rfloor$ points: zero out the indices with the most negative ψ_n. Re-fitting once gives an exact lower bound on true sensitivity.

9 Key Implications

- AMIP non-robustness does *not* vanish as $N \to \infty$ (unlike SEs).
- Not caused by misspecification or outliers - can occur in correctly-specified models.
- Distinct from gross-error robustness, trimming outliers does not fix it.

10 Applications

- **Oregon Medicaid:** dropping $< 1\%$ of $\sim$22K obs flips most health effects despite tiny p-values.
- **Progresa cash transfers:** non-poor effects flip with 3 obs — already-trimmed outliers don't help.
- **Microcredit (Mexico):** *one* observation flips the sign of the ATE.
- **Bayesian hierarchical microcredit:** Bayesian regularization doesn't fix it.

References

Anders Aamand, Justin Y. Chen, Huy Nguyen, Sandeep Silwal, and Ali Vakilian. Improved frequency estimation algorithms with and without predictions. In *Thirty-seventh Conference on Neural Information Processing Systems*, 2023. URL https://arxiv.org/abs/2312.07535.

Tamara Broderick, Ryan Giordano, and Rachael Meager. An automatic finite-sample robustness metric: When can dropping a little data make a big difference?, 2023. URL https://arxiv.org/abs/2011.14999.

Chen-Yu Hsu, Piotr Indyk, Dina Katabi, and Ali Vakilian. Learning-based frequency estimation algorithms. In *International Conference on Learning Representations*, 2019. URL https://openreview.net/forum?id=r1lohoCqY7.